

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 13: Traffic Alert & Collision Avoidance (TCAS II v7.1)

Syllabus Source: CAE Oxford (pp. 489-538) | Pass Target: >= 90% AviationExam

1. TCAS II ARCHITECTURE & THE TAU (TAU) PRINCIPLE

TCAS II operates independently of ground Air Traffic Control radar, interrogating SSR Mode C and Mode S transponders on 1030 MHz and receiving replies on 1090 MHz within a ~30 to 50 NM surveillance range:

- Tau Concept (Time to CPA): TCAS does NOT calculate collision risk based purely on physical distance! Instead, it evaluates TAU (Tau): Time to Closest Point of Approach (CPA): $\text{Tau} = \text{Range} / \text{Closure Rate}$: $\text{Tau} = r / (-r_{\text{dot}})$.
- Threat Classification Hierarchy: 1. Non-Threat Traffic: Open white diamond (> 6 NM or > 1,200 ft vertical); 2. Proximity Traffic: Solid white diamond (within 6 NM and 1,200 ft, but non-converging); 3. Traffic Advisory (TA): Solid amber circle; 4. Resolution Advisory (RA): Solid red square.

Alert Level	Tau Time Threshold (Cruise)	Cockpit Visual & Synthetic Aural Warning	Mandatory Pilot Action
Alert Level	Tau Time Threshold (Cruise)	Cockpit Visual & Synthetic Aural Warning	Mandatory Pilot Action
Traffic Advisory (TA)	Tau = 35 to 48 Seconds	Solid AMBER circle on ND. Synthetic voice: 'TRAFFIC, TRAFFIC'.	Attempt visual acquisition of conflicting traffic. PREPARE for possible Resolution Advisory. DO NOT MANEUVER solely based on TA!
Resolution Advisory (RA)	Tau = 20 to 35 Seconds	Solid RED square on ND. Pitch cues on PFD vertical speed tape (green fly-to arc; red avoid arc). Aural: 'CLIMB', 'DESCEND', 'LEVEL OFF'.	FLY THE RA IMMEDIATELY within 5 SECONDS! TCAS RA commands take strict priority over all ATC clearances!

2. TCAS II VERSION 7.1 ENHANCEMENTS & SENSE REVERSALS

Following the fatal 2002 Überlingen mid-air collision, TCAS Version 7.1 introduced vital safety modifications:

- 'LEVEL OFF' Aural Replaces 'Adjust Vertical Speed': In initial RA encounters where traffic levels off, TCAS v7.1 now commands 'LEVEL OFF, LEVEL OFF', instructing pilots to reduce vertical speed to zero fpm smoothly, preventing excessive climb/descent.
- Coordinated Sense Reversals: When two TCAS II aircraft encounter each other, their Mode S transponders negotiate resolution senses automatically: one commands CLIMB while the other commands DESCEND. If one pilot fails to comply (flying opposite to RA), TCAS v7.1 detects the non-compliance and issues an immediate SENSE REVERSAL ('CLIMB, CLIMB NOW!' or 'DESCEND, DESCEND NOW!'). Reaction time for reversal RA: 2.5 SECONDS!

[!] EXAM TRAP (AVIATIONEXAM) - ATC vs TCAS RA Absolute Legal Priority

- STRICT EASA / ICAO LAW: A TCAS Resolution Advisory (RA) ALWAYS TAKES PRECEDENCE over conflicting ATC instructions! - If ATC says 'Descend FL 350' but TCAS commands 'CLIMB, CLIMB': - YOU MUST CLIMB! Never follow an ATC clearance that contradicts a TCAS RA! - Notify ATC as soon as workload permits: '[Callsign], TCAS RA'.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): TCAS II Resolution Advisory Acti**

Rule 1: Disconnect autopilot and execute RA within 5 SECONDS applying 0.25g. Rule 2: OBEY TCAS OVER ATC: If controller orders opposite maneuver, pilot MUST follow TCAS RA. Rule 3: TCAS II commands ONLY VERTICAL maneuvers (Climb/Descend). It NEVER commands turns. Rule 4: Call ATC: '[Callsign], TCAS RA'.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: If an air traffic controller issues a radar descent clearance that directly contradicts a TCAS Resolution Adv**

- [A]: Follow the ATC instruction because ATC has radar separation overview
- [B]: Follow the TCAS RA immediately, disengaging autopilot if necessary, and advise ATC as soon as practical
- [C]: Level off and ask ATC to confirm
- [D]: Turn 90 degrees right

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under international aviation regulations, a TCAS RA takes absolute legal precedence over conflicting ATC clearances. Pilots must immediately respond to the RA within 5 seconds and notify ATC.

Q2: How does TCAS II calculate the time to Closest Point of Approach (CPA) to trigger a Traffic Advisory or Re

- [A]: By measuring barometric pressure difference
- [B]: Using the Tau (Tau) formula: Range divided by Closure Rate (r / r_{dot})
- [C]: By checking the GPS ground distance
- [D]: Using the aircraft Mach number

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. TCAS II evaluates collision risk using Tau (time to CPA), which is calculated as instantaneous slant range divided by closing velocity. This accounts for high closing speeds in head-on traffic.

Q3: What critical safety change was implemented in TCAS II Version 7.1 to improve pilot compliance during ve

- [A]: Added horizontal steering commands
- [B]: Replaced 'Adjust Vertical Speed, Adjust' with the clear synthetic voice command 'LEVEL OFF, LEVEL OFF'
- [C]: Increased surveillance range to 200 NM
- [D]: Inhibited all warnings below 10,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. TCAS v7.1 replaced the confusing 'Adjust Vertical Speed' command with 'LEVEL OFF, LEVEL OFF' to guide pilots smoothly to zero vertical speed, preventing overshoot into conflicting flight levels.